

PROJECT 03

Line Follower Robot with Gripper Arm

Effective Engineering & Automation (Client: School)
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MUDITHA PRIYASAD

Mechanical & Automation Engineer
BSc (Hons) Mech. Eng. — Univ. of Jaffna
SOLIDWORKS CSWA · CSWA-AM · IESL
AM-32795
muditha00@icloud.com · +94 71 7599 667

PROJECT SUMMARY

Designed, built, and programmed a line-following robot with a servo-driven gripper arm for use as an educational robotics demonstration at a school. The robot follows a black line, picks up 40 mm wooden cubes, detects object colour, and places them at target locations. All parts were 3D-printed in-house.

OBJECTIVES

- Keep robot within 150 mm wide × 210 mm long.
- Follow a black line automatically using PID steering control.
- Pick up and place 40×40×40 mm wooden cubes (max 50 g).
- Detect and identify object colour using an RGB sensor.

HOW IT WORKS

Differential drive (two DC motors, one per rear wheel) with PID-controlled steering from a 5-channel IR sensor array. A 2-DoF crane arm at the front uses servo motors for vertical lift and gripper open/close. A TCS3200 colour sensor on the gripper reads object colour before pick-up. Arduino Uno runs all logic. All structural parts are 3D-printed PLA.

KEY COMPONENTS

Component	Specification / Function
Arduino Uno	Main microcontroller
IR Sensor Array (5ch)	Line detection
TCS3200 Colour Sensor	Object colour classification
L298N Motor Driver	Controls two DC drive motors
DC Gear Motor ×2	6V, 150 RPM — differential drive
9g Servo Motor ×2	Arm lift and gripper actuation
7.4V LiPo Battery	Power supply (motors + logic)
3D-Printed Parts	PLA chassis, arm, and gripper housing

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
Footprint	148 × 206 mm	Within 150 × 210 mm limit
Steering control	PID algorithm	Kp/Ki/Kd tuned empirically
Arm DoF	2 (lift + gripper)	—
Max payload	50 g cube (40 mm)	—
Microcontroller	Arduino Uno, 16 MHz	—
Structure	PLA, FDM 3D-printed	—

MY CONTRIBUTION

- Designed chassis, arm, and gripper in SolidWorks; printed all parts in-house.
- Assembled and wired all electronics.

- Programmed PID line-following and arm/colour-sensor logic in Arduino C++.
- Tuned PID gains on test track; validated pick-and-place over 20 cycles.
- Delivered working robot to client school.

CHALLENGES & SOLUTIONS

PID oscillated at higher speeds — fixed by reducing derivative gain and limiting top speed. Gripper alignment needed three CAD revisions. Colour sensor affected by ambient light — solved by adding a sensor shroud.

RESULTS

Robot tracks line at 0.25 m/s. 95% pick-up success over 20 trials. Correct colour detection achieved. All size limits met. Delivered as a working school demonstration.

TOOLS & SOFTWARE USED

Tool / Software	Used For
SolidWorks	Chassis, arm, and gripper design
Arduino IDE	PID, arm control, and colour detection code
Cura	Slicer for 3D-printed parts
Custom 3D Printer	Printing all structural parts
Multimeter	Electronics verification

IMAGES & TECHNICAL DRAWINGS

[Insert SolidWorks renders, fabrication drawings, assembly photos here.]